IDH 313 – MIKROTIC Certified Network Training

MikroTik Certified Network Associate (MTCNA) – Bridging

MikroTik MTCNA – Introduction and RouterOS Fundamentals

Module Title: MikroTik Fundamentals and First-Time Router Setup

Duration: 6–8 hours (including hands-on labs)
Prerequisites: Basic networking knowledge (IP addresses, LAN/WAN concepts).

Learning Objectives:

- **Understand what MikroTik, RouterBOARD, and RouterOS are.**
- **Access and configure a router for the first time.**
- **Use WinBox, WebFig, CLI, SSH, and Telnet for router management.**
- **Perform initial Internet configuration and basic firewall setup.**
- **Upgrade RouterOS and manage backups.**
- **Understand RouterOS license levels and support resources.**

1. About MikroTik

1.1 What is RouterOS?

- **RouterOS is MikroTik's operating system for routers.**
- **Features: Routing, Firewall, NAT, VPN, Wireless, QoS, Monitoring.**
- **Can be installed on RouterBOARD or standard PC hardware.**

1.2 What is RouterBOARD?

- **MikroTik's hardware platform for RouterOS.**
- **Features:**
    - **Ethernet ports (LAN/WAN)**
    - **CPU, RAM, Flash storage**

- Wireless interfaces (depending on model)
- RouterBOOT for firmware/boot management

2. First-Time Router Access

2.1 Access Methods

- WinBox: GUI management tool (IP or MAC address).
- MAC-WinBox: Connect using MAC address when IP is unknown.
- WebFig: Browser-based configuration.
- Quick Set: Simplified setup for first-time configuration.
- CLI / Terminal: For command-line configuration.

Lab Exercise:

- Connect PC to RouterBOARD via Ethernet.
- Launch WinBox and connect using MAC address.
- Access the router via WebFig in a browser.

2.2 RouterOS CLI Principles

- Commands can be tab-completed (<tab>)
- ? shows available commands.
- Double <tab> lists all options for a command.
- Command history: Use up/down arrows to navigate previously entered commands.

CLI Example:

/ip address add address=192.168.88.2/24 interface=ether2

3. Initial Router Configuration

3.1 WAN DHCP Client

- Configure router to get IP from ISP dynamically.

/ip dhcp-client add interface=ether1 disabled=no

3.2 LAN IP and Default Gateway

/ip address add address=192.168.88.1/24 interface=ether2

/ip route add gateway=192.168.88.254

3.3 Basic Firewall NAT (Masquerade)

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade

4. Access Methods

Method	Use Case	Notes
WinBox	GUI, full router configuration	MAC or IP login
WebFig	Browser GUI	Limited functionality, quick setup
CLI	Scriptable, advanced tasks	Terminal or SSH/Telnet
Null Modem Cable	Direct serial access	For routers without IP access
SSH	Remote secure command-line	Requires SSH server enabled
Telnet	Remote unsecure command-line	Not recommended for WAN

Lab Exercise:

- **Access router via CLI in WinBox terminal.**
- **Experiment with <tab> and ? for command discovery.**

5. Upgrading RouterOS

5.1 Package Types

- **Main Packages: Base system, routing, firewall, wireless.**
- **Optional Packages: Additional features like User Manager.**

5.2 Ways of Upgrading

- **Via WinBox/WebFig → System → Packages → Check for Updates**
- **Manually upload .npk files via FTP/WinBox.**

5.3 RouterBOOT Firmware Upgrade

- **Update RouterBOOT to fix boot issues or enable new features.**
- **CLI Example:**

/system routerboard upgrade

6. Router Identity and Management

6.1 Router Identity

- **Set a unique name for router identification:**

/system identity set name="Office-Router"

6.2 RouterOS Logins

- **Default user: admin (no password initially)**
- **Add users and assign groups/permissions:**

/user add name="tech" group=full password="tech123"

6.3 Managing Services

- **Enable/disable services (WinBox, SSH, HTTP):**

/ip service print

/ip service disable telnet

7. Configuration Backups

7.1 Backup

- **Saves router in binary format (can restore to same or similar router).**

/system backup save name=backup1

/system backup load name=backup1

7.2 Export

- **Saves configuration as readable .rsc file.**

/export file=config1

/import file=config1.rsc

7.3 Editing Export File

- **.rsc files are text-based and can be edited before import.**
- **Useful for scripting multiple router configurations.**

8. Resetting and Reinstalling RouterOS

8.1 Resetting Router

- **Restore default settings via WinBox/WebFig or CLI:**

/system reset-configuration

8.2 Reinstalling via Netinstall

- **Boot router in Netinstall mode using PC to reinstall RouterOS completely.**
- **Useful for corrupted systems or changing RouterOS versions.**

9. RouterOS License Levels

Level	Max Interfaces	Notes
3	3	Basic use
4	5	Small office
5	10	Medium office
6	10 +	Unlimited, full features
0	Demo	Limited features, expiration

10. Additional Information Sources

- MikroTik Wiki → Documentation
- MikroTik Forum → Community support
- MUM – MikroTik User Meetings → Training/events
- Distributor / consultant support
- Email: support@mikrotik.com

11. Laboratory Exercises

Lab 1: First-Time Router Access

- Connect router via Ethernet.
- Access router using MAC WinBox and WebFig.
- Explore Quick Set interface.

Lab 2: Basic CLI Navigation

- Use <tab>, ?, and command history.
- Explore interfaces, IP addresses, and firewall rules.

Lab 3: Internet Access

- Configure WAN DHCP client.
- Assign LAN IP and gateway.
- Test Internet connectivity.

Lab 4: Basic Firewall & NAT

- Apply NAT masquerade for Internet sharing.
- Test LAN devices accessing the Internet.

Lab 5: Backup & Restore

- Save backup and export configuration.
- Reset router and restore configuration from backup and .rsc file.

Lab 6: RouterOS Upgrade

- Check RouterOS version.

- **Upgrade using WinBox/WebFig.**
- **Upgrade RouterBOOT firmware.**

12. Learning Outcomes

After this module, students will be able to:

1. **Explain RouterOS and RouterBOARD fundamentals.**
2. **Access routers using WinBox, WebFig, CLI, SSH, Telnet, and MAC WinBox.**
3. **Perform initial router configuration for Internet access.**
4. **Apply basic firewall and NAT rules.**
5. **Backup, restore, and upgrade RouterOS.**
6. **Manage users, services, and Router identity.**
7. **Use RouterOS documentation and support resources efficiently.**

Chapter Two

MikroTik MTCNA – DHCP Server and Client Module

Module Title: DHCP and ARP in MikroTik RouterOS

Duration: 4–6 hours (including hands-on labs)
Prerequisites: Basic knowledge of IP addressing, LAN configuration, and RouterOS fundamentals.

Learning Objectives:

- **Understand DHCP concepts and RouterOS DHCP implementation.**
- **Configure DHCP client and server on MikroTik routers.**
- **Manage DHCP leases and networks.**
- **Understand ARP and ARP modes for IP-to-MAC resolution.**
- **Use RouterOS ARP table for monitoring and troubleshooting.**

1. DHCP Concepts

1.1 What is DHCP?

- **Dynamic Host Configuration Protocol (DHCP) dynamically assigns IP addresses and other network parameters (gateway, DNS) to clients.**
- **Reduces manual configuration errors and simplifies network management.**

1.2 DHCP Client

- **Device that requests IP and network configuration from a DHCP server.**
- **In RouterOS, WAN interfaces often use DHCP client to get IP from ISP.**

CLI Example:

/ip dhcp-client add interface=ether1 disabled=no

/ip dhcp-client print

WinBox Steps:

1. **IP → DHCP Client → Add.**

2. **Select interface (e.g., ether1).**
3. **Enable and apply.**

1.3 DHCP Server

- **Assigns IP addresses to clients in a defined network.**
- **Components:**
    - **IP Pool: Range of assignable IPs.**
    - **Network: IP range, gateway, DNS servers.**
    - **Lease Time: Duration an IP is assigned.**

CLI Example:

```
# Define IP pool

/ip pool add name=dhcp_pool ranges=192.168.88.10-192.168.88.50


# Define DHCP network

/ip dhcp-server network add address=192.168.88.0/24 gateway=192.168.88.1 dns-server=8.8.8.8


# Enable DHCP server

/ip dhcp-server add name=dhcp1 interface=ether2 address-pool=dhcp_pool disabled=no

/ip dhcp-server enable dhcp1
```

WinBox Steps:

1. **IP → DHCP Server → DHCP Setup.**
2. **Select interface, pool, network, and gateway.**
3. **Complete setup and activate DHCP server.**

2. DHCP Leases Management

- **Leases: Record of IP addresses assigned to clients.**
- **Types: Active, Static, Expired.**
- **Manage leases via CLI or WinBox.**

CLI Examples:

/ip dhcp-server lease print

/ip dhcp-server lease add address=192.168.88.20 mac-address=00:11:22:33:44:55 comment="Printer"

WinBox Steps:

- **IP → DHCP Server → Leases → Add/Remove/Make Static.**

3. DHCP Server Network Configuration

- **Network Parameters:**
  - **Gateway**
  - **DNS Server(s)**
  - **Domain name**
  - **NTP server (optional)**

CLI Example:

/ip dhcp-server network set 0 gateway=192.168.88.1 dns-server=8.8.8.8,8.8.4.4

- **Ensures clients automatically receive correct network settings.**

4. Address Resolution Protocol (ARP)

4.1 What is ARP?

- **ARP maps IP addresses to MAC addresses on a LAN.**
- **Essential for Ethernet communication.**

4.2 ARP Table in RouterOS

CLI Example:

/ip arp print

/ip arp add address=192.168.88.100 mac-address=00:11:22:33:44:66 interface=ether2

- **Shows current IP-MAC mappings and interfaces.**

4.3 ARP Modes

- **enabled: Normal ARP behavior (default).**
- **disabled: No ARP entries, only manually defined static entries work.**
- **proxy-arp: Router answers ARP requests on behalf of other devices. Useful for bridging or VPN.**

CLI Example:

/interface ethernet set ether2 arp=proxy-arp

5. Laboratory Exercises

Lab 1: DHCP Client Configuration

- **Configure WAN interface as DHCP client.**
- **Verify IP assignment using /ip dhcp-client print.**
- **Test Internet connectivity using ping 8.8.8.8.**

Lab 2: DHCP Server Setup

- **Create IP pool and network.**
- **Enable DHCP server on LAN interface.**
- **Connect a client PC and verify automatic IP assignment.**

Lab 3: DHCP Lease Management

- **List active leases.**
- **Reserve a static IP for a client device (e.g., printer).**
- **Release and renew a lease from a client PC.**

Lab 4: ARP Monitoring

- View ARP table.
- Add static ARP entries.
- Change ARP mode to proxy-arp and test traffic forwarding.

Lab 5: DHCP + Firewall Integration

- Combine DHCP server with basic firewall (masquerade).
- Test LAN client Internet access.

6. Learning Outcomes

By the end of this module, students will be able to:

1. Explain DHCP concepts and its role in dynamic IP assignment.
2. Configure RouterOS DHCP client and server.
3. Manage DHCP leases and assign static IPs.
4. Configure DHCP server network parameters (gateway, DNS).
5. Understand ARP, view ARP table, and configure ARP modes.
6. Integrate DHCP with basic router firewall for Internet access.

Chapter Three

Module Title: Bridging in MikroTik RouterOS

Duration: 6–8 hours (including lab exercises)
Prerequisites: Basic knowledge of MikroTik RouterOS interface, IP addressing, and networking concepts.

Learning Objectives:

- Understand bridging concepts in Layer 2 networks.
- Configure and manage bridges in RouterOS.
- Integrate wired and wireless networks using bridges.
- Implement Station Bridge for point-to-point wireless links.
- Troubleshoot bridge-related network issues.

1. Bridging Overview

1.1 What is a Bridge?

A **bridge** is a Layer 2 device or interface that connects multiple network segments to function as a single network. It forwards traffic based on MAC addresses, not IP addresses. In MikroTik RouterOS, bridges allow you to **combine multiple interfaces** (Ethernet or wireless) to act as one LAN segment.

Benefits of Bridging:

- Expands network without additional routers.
- Integrates wired and wireless networks.
- Supports VLANs (Virtual LANs) for network segmentation.
- Simplifies management of multiple interfaces.
- Supports Spanning Tree Protocol (STP/RSTP) for loop prevention.

Layer Comparison:

Layer	Device	Decision based on
L2	Bridge	MAC Address

Layer	Device	Decision based on
L3	Router	IP Address

1.2 Bridge vs Switch

- RouterOS bridge acts like a software switch.
- Forwarding is based on MAC addresses.
- Can include wireless interfaces, VLANs, or physical ports.
- Supports STP/RSTP to prevent loops, unlike a simple unmanaged switch.

Example Use Case:
You have a MikroTik router with 4 Ethernet ports and 1 wireless AP. You can bridge all ports and the wireless interface to create a single LAN.

2. Bridge Concepts and Settings

2.1 Bridge Interface in RouterOS

A bridge interface acts as a logical entity. You can add physical or virtual ports to it.
Bridge settings include:

- **Forwarding:** Enable/disable packet forwarding between ports.
- **STP / RSTP:** Loop prevention protocols.
- **Priority and Path Cost:** STP configuration.
- **Fast Forwarding:** Optimizes performance for low latency.
- **Hairpin Mode:** Allows NAT traffic to loop back within a bridge.
- **MAC Address Table:** Maintains a list of all connected MAC addresses.

Command Example:

```
/interface bridge add name=BR-LAN

/interface bridge port add bridge=BR-LAN interface=ether2

/interface bridge port add bridge=BR-LAN interface=ether3
```

2.2 STP / RSTP

- **STP (Spanning Tree Protocol):** Prevents loops in Layer 2 networks.
- **RSTP (Rapid STP):** Faster convergence.
- **Port States:** Disabled, Blocking, Listening, Learning, Forwarding.

Best Practice: Use **RSTP** in modern networks for faster convergence.

2.3 Bridge MAC Address Table

- Shows which MAC addresses are reachable through which bridge port.
- Helps identify which devices are on each interface.
- Commands:

/interface bridge host print

3. Creating Bridges

3.1 Using WinBox

1. Open WinBox → Go to **Bridge → Add (+)**.
2. Name the bridge (e.g., BR-LAN).
3. Keep default settings (RSTP enabled).
4. Click **Apply** → **OK**.

3.2 Using CLI

/interface bridge add name=BR-LAN

Verify Bridge:

/interface bridge print

4. Adding Ports to Bridges

4.1 Adding Physical Ports

- Each Ethernet port can be added to a bridge as a **bridge port**.

- Commands:

/interface bridge port add bridge=BR-LAN interface=ether2

/interface bridge port add bridge=BR-LAN interface=ether3

4.2 Port Settings

- **PVID:** VLAN ID assigned to untagged packets on this port.
- **Ingress Filtering:** Drop packets that don't belong to allowed VLANs.
- **Frame Types:** Admit only tagged, untagged, or all frames.

Lab Exercise:

- Add ports ether2 and ether3 to bridge BR-LAN.
- Assign PVID 1 for default VLAN.
- Connect devices and verify connectivity using ping.

5. Bridging Wireless Networks

5.1 AP Bridge Mode

- Used to act as a wireless Access Point.
- Can be added to a bridge to extend the LAN.

Configuration Example:

/interface wireless set wlan1 mode=ap-bridge ssid="MikroTik-WiFi"

/interface bridge port add bridge=BR-LAN interface=wlan1

5.2 Repeater Mode

- Extends the wireless network range.
- Can also be part of a bridge.

Lab Exercise:

- Bridge wlan1 to BR-LAN.
- Connect a wireless client.
- Ping wired devices in the LAN.

6. Station Bridge

6.1 Concept

- **Station:** Connects to AP, manages router interface only.
- **Station Bridge:** Connects to AP Bridge and extends Layer 2 network transparently.

6.2 Use Case

- Point-to-point wireless link.
- Extending LAN across buildings without additional routers.

6.3 Configuration Example

On the client router (Station Bridge):

```
/interface wireless set wlan1 mode=station-bridge ssid="MikroTik-WiFi"
/interface bridge add name=BR-LAN
/interface bridge port add bridge=BR-LAN interface=wlan1
/interface bridge port add bridge=BR-LAN interface=ether2
```

Lab Exercise:

- Router A: AP Bridge → SSID: MikroTik-WiFi.
- Router B: Station Bridge → Join Router A.
- Add Station Bridge interface + ether2 to bridge on Router B.
- Test connectivity between LAN devices across routers.

7. Module 3 Laboratory Exercises

Lab 1: Create a Bridge

- Create BR-LAB in WinBox or CLI.
- Verify bridge creation.
- Expected Outcome: Bridge appears in interface list.

Lab 2: Add Ethernet Ports

- Add ether2 and ether3 to BR-LAB.
- Verify MAC address table:

/interface bridge host print

- Test ping between connected devices.

Lab 3: Bridge Wireless AP

- Add wlan1 AP Bridge to BR-LAB.
- Connect wireless client → Verify ping to LAN.

Lab 4: Station Bridge Configuration

- Router A: AP Bridge with SSID Bridge-WiFi.
- Router B: Station Bridge → Join Router A.
- Add wlan1 and ether2 to bridge.
- Test connectivity → Ping, file transfer.

Lab 5: STP/RSTP Testing

- Enable RSTP on bridge.

/interface bridge set BR-LAB protocol-mode=rstp

- Connect extra cable to simulate loop.
- Observe port states → Forwarding/Blocking.

Lab 6: Monitoring and Troubleshooting

- Use Torch to monitor traffic:

/tool torch interface=BR-LAN

- Verify bridge MAC table for connected devices.
- Identify and fix any connectivity issues.

Learning Outcomes

After completing this module, students will be able to:

1. Explain bridging and Layer 2 networking principles.

2. Configure and manage bridges using CLI and WinBox.
3. Add Ethernet and wireless interfaces to bridges.
4. Implement Station Bridge for point-to-point wireless links.
5. Enable STP/RSTP and prevent loops in Layer 2 networks.
6. Monitor traffic and troubleshoot bridge connectivity issues.

Routing Module

Module Title: Routing in MikroTik RouterOS

Duration: 6–8 hours (including lab exercises)
Prerequisites: Basic knowledge of IP addressing, MikroTik interfaces, and Layer 2 connectivity (bridges).

Learning Objectives:

- Understand routing concepts and route types in MikroTik RouterOS.
- Configure static and dynamic routes.
- Set default routes for Internet access.
- Understand and manage route flags.
- Implement routing in a small network using static routes.
- Monitor and troubleshoot routes effectively.

1. Routing Overview

1.1 What is Routing?

Routing is the process of **determining the path packets take to reach a destination network**. It operates at **Layer 3 (Network Layer)** of the OSI model.

- **Router Role:** Connects multiple networks, decides where to forward packets based on destination IP.
- **Routing Table:** Stores available routes and associated next-hop information.
- **Forwarding:** Routes can be **static** (manually configured) or **dynamic** (learned automatically via protocols).

RouterOS Routing Tables:

- /ip route print → Displays all routes.
- Attributes: Destination, Gateway, Distance, Scope, Flags.

1.2 Routing Concepts

- **Destination Network:** The network to which packets are sent.
- **Gateway (Next Hop):** IP address of the next router to forward packets to the destination.
- **Interface:** Outgoing interface used to reach the gateway.
- **Distance / Metric:** Preference of a route. Lower distance = higher priority.

Example:

Destination: 192.168.2.0/24

Gateway: 192.168.1.2

Interface: ether2

Distance: 1

2. Route Flags

RouterOS uses **route flags** to indicate route type and status. Some important flags:

Flag	Description
D	Dynamic (learned via routing protocol)
S	Static (manually configured)
C	Connected (directly attached network)
A	Active route (currently used)
!	Invalid route (next hop unreachable)
B	Blackhole route

Command to check flags:

/ip route print

3. Static Routing

3.1 What is Static Routing?

- Manually defined routes by network administrator.
- Used for **small networks** or where predictable routing is required.
- Advantages:
  - Simple and easy to control.
  - No routing protocol overhead.
- Disadvantages:
  - Manual maintenance required.
  - Not scalable for large networks.

4. Creating Routes

4.1 Using CLI

Syntax:

/ip route add dst-address=<destination_network> gateway=<gateway_ip> distance=<value>

Example:

/ip route add dst-address=192.168.2.0/24 gateway=192.168.1.2 distance=1

4.2 Using WinBox

1. Open **IP → Routes → Add (+)**.
2. Destination: 192.168.2.0/24.
3. Gateway: 192.168.1.2.
4. Distance: 1.
5. Click **Apply → OK**.

Verify Route:

/ip route print

5. Setting Default Route

A **default route** is used when no specific route exists for a destination (often used for Internet access).

5.1 CLI Example

/ip route add dst-address=0.0.0.0/0 gateway=192.168.1.1

5.2 WinBox Example

1. IP → Routes → Add (+)
2. Destination: 0.0.0.0/0
3. Gateway: 192.168.1.1
4. Apply and OK.

Test:

ping 8.8.8.8

Should be reachable via default route.

6. Managing Dynamic Routes

RouterOS supports dynamic routing protocols:

- **RIP:** Suitable for small networks.
- **OSPF:** Large LANs.
- **BGP:** Wide-area networks (Internet connections).

Viewing dynamic routes:

/ip route print where dynamic

Lab Focus:
In MTCNA, emphasis is on **static routes**, but students should understand that dynamic routes are automatically learned and flagged D.

7. Implementing Static Routing in a Simple Network

7.1 Network Scenario

- Router A: LAN 192.168.1.0/24, ether1 to Internet, ether2 to Router B.
- Router B: LAN 192.168.2.0/24, ether1 to Router A, ether2 to PCs.
- Objective: Enable PCs in 192.168.2.0/24 to reach Internet via Router A.

7.2 Configuration Steps

Step 1: Configure LAN interfaces

```
# Router A
/ip address add address=192.168.1.1/24 interface=ether2

# Router B
/ip address add address=192.168.2.1/24 interface=ether2
/ip address add address=192.168.1.2/24 interface=ether1
```

Step 2: Configure static routes

```
# Router B: route to Internet via Router A
/ip route add dst-address=0.0.0.0/0 gateway=192.168.1.1
```

Step 3: Test connectivity

- From Router B LAN: ping 8.8.8.8
- From Router A LAN: ping 192.168.2.1

8. Laboratory Exercises

Lab 1: Adding Static Routes

- Add route 192.168.3.0/24 via gateway 192.168.2.1.
- Verify connectivity.

Lab 2: Setting Default Route

- Configure default route to Internet gateway.
- Test connectivity using ping and traceroute.

Lab 3: Multi-LAN Routing

- Scenario: Router with 3 LANs.
- Add static routes between LANs.
- Test inter-LAN connectivity.

Lab 4: Observing Route Flags

- Use /ip route print to view connected, static, and active routes.
- Identify flags: C for connected, S for static, A for active.

Lab 5: Troubleshooting Routing

- Remove a gateway temporarily.
- Observe route flags (! for invalid route).
- Restore gateway and verify.

Learning Outcomes

By the end of this module, students will be able to:

1. Explain routing concepts and Layer 3 forwarding.
2. Identify and interpret route flags in MikroTik RouterOS.
3. Configure static routes for specific networks.
4. Set a default route for Internet access.
5. Implement routing between multiple LANs.
6. Monitor and troubleshoot routes using CLI and WinBox.

MikroTik MTCNA – Wireless Module

Module Title: Wireless Networking in MikroTik RouterOS

Duration: 6–8 hours (including lab exercises)
Prerequisites: Basic knowledge of RouterOS interfaces, IP addressing, and bridging concepts.

Learning Objectives:

- Understand wireless standards and frequency concepts.
- Configure MikroTik routers as Access Points and Stations.
- Implement wireless security and access control.
- Monitor wireless performance and client activity using RouterOS tools.

1. Wireless Standards and Concepts

1.1 802.11a/b/g/n/ac Overview

Standard	Frequency	Max Data Rate	Notes
802.11a	5 GHz	54 Mbps	Shorter range, less interference
802.11b	2.4 GHz	11 Mbps	Longer range, more interference
802.11g	2.4 GHz	54 Mbps	Backward compatible with 802.11b
802.11n	2.4/5 GHz	600 Mbps	MIMO support (multiple antennas)
802.11ac	5 GHz	1.3 Gbps+	MU-MIMO, higher throughput

Key Terms:

- **SSIDs:** Network identifiers.
- **MIMO Chains:** Number of transmit/receive antennas (1x1, 2x2, 3x3...).
- **TX Power:** Transmission strength (higher = longer range, more interference).

- **RX Sensitivity:** Minimum signal level the receiver can handle.

1.2 Frequency Bands and Channels

- **2.4 GHz band:** Channels 1–13 (depends on country).
    - Pros: Longer range, better penetration.
    - Cons: Crowded, more interference.
- **5 GHz band:** Channels 36–165.
    - Pros: Less interference, higher throughput.
    - Cons: Shorter range, stricter country regulations.

RouterOS Setting Example:

```
/interface wireless set wlan1 frequency=2412 band=2ghz-b/g/n
```

Country Regulations:

- Ensure TX power and channel comply with local regulations.
- RouterOS uses **regulatory domain** to enforce this.

```
/interface wireless set wlan1 country="united states"
```

2. Wireless Link Setup

2.1 Simple Wireless Link

Scenario: Point-to-point connection between two MikroTik routers.

Steps:

1. Router A: AP Bridge
2. Router B: Station
3. Set SSID, frequency, and wireless mode.

Example CLI:

```
# Router A (AP Bridge)
interface wireless set wlan1 mode=ap-bridge ssid="MikroLink"
```

interface bridge port add bridge=BR-LAN interface=wlan1

Router B (Station)

interface wireless set wlan1 mode=station ssid="MikroLink"

interface bridge port add bridge=BR-LAN interface=wlan1

Lab Exercise:

- Create a simple wireless link between two routers.
- Test connectivity with ping between LAN devices on both sides.

2.2 Access Point Configuration

Key Settings:

- **Mode:** AP Bridge
- **SSID:** Network name
- **Frequency & Band**: 2.4GHz or 5GHz
- **Security Profile:** WPA/WPA2

CLI Example:

/interface wireless set wlan1 mode=ap-bridge ssid="MikroAP" band=2ghz-b/g/n

/interface wireless security-profiles add name=SecProfile authentication-types=wpa2-psk wpa2-pre-shared-key="mikrotik123"

/interface wireless set wlan1 security-profile=SecProfile

WinBox Steps:

1. Wireless → Interface → wlan1 → Mode = AP Bridge.
2. Set SSID, Frequency, Band.
3. Assign Security Profile.

2.3 Station Configuration

- **Station Mode:** Connects to AP, only allows router LAN access.
- **Station Bridge Mode:** Extends AP LAN over wireless (Layer 2).

CLI Example:

/interface wireless set wlan1 mode=station-bridge ssid="MikroAP"

/interface bridge port add bridge=BR-LAN interface=wlan1

3. Wireless Security and Encryption

3.1 Security Methods

- **WPA-PSK:** Wi-Fi Protected Access – Pre-shared Key
- **WPA2-PSK:** Stronger encryption (AES recommended)
- **WPS (Wi-Fi Protected Setup):** Accept / Client mode

CLI Example (WPA2-PSK):

/interface wireless security-profiles add name=WPA2-Profile authentication-types=wpa2-psk wpa2-pre-shared-key="MikroKey123"

/interface wireless set wlan1 security-profile=WPA2-Profile

3.2 Access Control

- **Access List:** Define allowed MAC addresses.
- **Connect List:** Automatic connections based on MAC.
- **Default Authenticate:** Whether unknown clients are allowed.
- **Default Forward:** Allow bridged traffic to LAN.

CLI Example:

/interface wireless access-list add interface=wlan1 mac-address=00:11:22:33:44:55 forward=yes comment="Allowed Client"

/interface wireless set wlan1 default-authentication=yes default-forward=yes

4. Monitoring Tools

4.1 Snooper

- Tool to scan wireless spectrum for networks.
- CLI: /interface wireless snooper set interface=wlan1
- Shows: SSID, Signal Strength, Channel, Frequency.

4.2 Registration Table

- Lists all clients connected to AP.
- CLI: /interface wireless registration-table print
- Shows: MAC, IP, Signal Strength, Tx/Rx rate, uptime.

5. Laboratory Exercises

Lab 1: Basic Wireless AP

- Configure wlan1 as AP Bridge.
- Set SSID: LabWiFi.
- Apply WPA2-PSK encryption.
- Connect a client device and test LAN access.

Lab 2: Station / Station Bridge

- Router A: AP Bridge LabWiFi.
- Router B: Station Bridge → join AP.
- Add wlan1 to bridge on Router B.
- Verify connectivity between LANs.

Lab 3: Access Control

- Add MAC addresses to Access List.
- Test allowed and denied clients.

Lab 4: WPA2 Security

- Configure WPA2-PSK on AP.
- Connect a wireless client using pre-shared key.
- Verify connectivity and encryption.

Lab 5: Wireless Monitoring

- Use **Snooper** to scan networks and channels.
- Use **Registration Table** to view connected clients.
- Adjust TX Power and channels to optimize performance.

6. Learning Outcomes

By the end of this module, students will be able to:

1. Describe wireless standards (802.11 a/b/g/n/ac) and MIMO concepts.
2. Configure basic wireless links in AP and Station modes.
3. Implement secure wireless networks using WPA/WPA2-PSK.
4. Control client access using Access Lists and Connect Lists.
5. Use monitoring tools to optimize wireless performance.
6. Troubleshoot wireless client connectivity and link issues.

MikroTik MTCNA – Firewall and NAT

Module Title: Firewall and NAT in MikroTik RouterOS

Duration: 6–8 hours (including lab exercises)
Prerequisites: Knowledge of RouterOS interfaces, IP addressing, bridging, routing, and wireless networks.

Learning Objectives:

- Understand firewall principles and connection tracking.
- Configure RouterOS firewall filter rules to protect routers and networks.
- Implement NAT (Source NAT and Destination NAT).
- Use FastTrack for optimized routing performance.
- Manage address lists for network protection and access control.

1. Firewall Principles

1.1 What is a Firewall?

A **firewall** is a security system that monitors and controls network traffic based on **rules**. It protects the router itself and the network it manages.

Key Points:

- Operates at **Layer 3 (IP)** and Layer 4 (TCP/UDP).
- Works with **connection tracking** to identify traffic states.
- Rules are processed in **chains** with **actions** (accept, drop, reject).

1.2 Connection Tracking and States

RouterOS tracks all active connections for proper firewall operation.
Connection States:

- **New:** First packet of a connection.
- **Established:** Part of an existing connection.

- **Related:** Related to an existing connection (e.g., FTP data channel).
- **Invalid:** Malformed or unrecognized packets.

CLI Example:

/ip firewall connection print

2. Firewall Structure, Chains, and Actions

2.1 Chains

MikroTik firewall rules are organized into **chains**:

Chain	Description
input	Traffic destined to the router itself
forward	Traffic passing through the router (LAN → LAN, LAN → WAN)
output	Traffic generated by the router
prerouting	NAT before routing decision
postrouting	NAT after routing decision

2.2 Actions

Common firewall actions include:

- **accept:** Allow traffic.
- **drop:** Silently discard traffic.
- **reject:** Discard traffic and send an error response.
- **log:** Log packet info.
- **mark:** Mark connection/packet for routing or QoS.

3. Firewall Filter in Action

3.1 Protecting Your Router (Input Chain)

- Goal: Block unwanted traffic to router services (WinBox, SSH, HTTP).
  CLI Example:

/ip firewall filter add chain=input connection-state=established,related action=accept

/ip firewall filter add chain=input connection-state=invalid action=drop

/ip firewall filter add chain=input protocol=tcp dst-port=8291 action=drop

3.2 Protecting Your Customers (Forward Chain)

- Goal: Block unwanted traffic through the router.
  CLI Example:

/ip firewall filter add chain=forward connection-state=established,related action=accept

/ip firewall filter add chain=forward connection-state=invalid action=drop

/ip firewall filter add chain=forward src-address=192.168.2.100 action=drop

4. Basic Address-List

- Address lists allow grouping IPs for easier firewall management.
  CLI Example:

/ip firewall address-list add list=blocked_ips address=192.168.2.50 comment="Blocked client"

/ip firewall filter add chain=forward src-address-list=blocked_ips action=drop

5. Source NAT (SNAT) and Masquerade

5.1 Source NAT

- Translates private IPs to public IP for Internet access.
  CLI Example:

/ip firewall nat add chain=srcnat src-address=192.168.1.0/24 action=src-nat to-addresses=203.0.113.10

5.2 Masquerade

- Dynamic source NAT for connections with dynamic public IP.
  CLI Example:

/ip firewall nat add chain=srcnat out-interface=ether1 action=masquerade

6. Destination NAT (DNAT) and Redirect

6.1 DNAT

- Redirect external traffic to internal servers.
  Example: Forward HTTP traffic to internal web server.

/ip firewall nat add chain=dstnat dst-address=203.0.113.10 protocol=tcp dst-port=80 action=dst-nat to-addresses=192.168.1.100 to-ports=80

6.2 Redirect

- Redirect traffic to router itself (e.g., transparent proxy, hotspot).
  Example:

/ip firewall nat add chain=dstnat dst-port=80 protocol=tcp action=redirect to-ports=8080

7. FastTrack

- FastTrack is a MikroTik feature to **accelerate packet forwarding** by bypassing full firewall processing for established connections.
- Improves throughput and reduces CPU load.

CLI Example:

/ip firewall filter add chain=forward connection-state=established,related action=fasttrack-connection

Notes:

- Cannot be combined with some advanced firewall or QoS rules.
- Use for trusted traffic only.

8. Laboratory Exercises

Lab 1: Protect the Router

- Configure input chain rules to:
  - Accept established/related connections.

    - Drop invalid connections.
    - Block WinBox from Internet.
- Test by attempting unauthorized connections.

Lab 2: Protect Customers

- Configure forward chain rules to block invalid connections.
- Drop traffic from a specific IP in LAN.
- Test with ping and browsing.

Lab 3: Address List

- Add IPs to blocked_ips list.
- Apply firewall rule to drop traffic from list.
- Test access from blocked IPs.

Lab 4: Source NAT and Masquerade

- Configure src-nat for LAN 192.168.1.0/24 to go through WAN IP.
- Use Masquerade for dynamic public IP.
- Test Internet connectivity from LAN devices.

Lab 5: Destination NAT

- Forward external HTTP (port 80) to internal web server 192.168.1.100.
- Test from external network.

Lab 6: FastTrack

- Enable FastTrack for established connections.
- Test performance improvement using bandwidth test tools.

9. Learning Outcomes

By the end of this module, students will be able to:

1. Explain firewall principles and connection tracking states.
2. Configure RouterOS firewall filters for router protection and network traffic control.
3. Create and manage address lists for easy access control.

4. Implement source NAT and destination NAT for Internet access and port forwarding.
5. Apply FastTrack to accelerate established connections.
6. Monitor and troubleshoot firewall rules and NAT translations effectively.